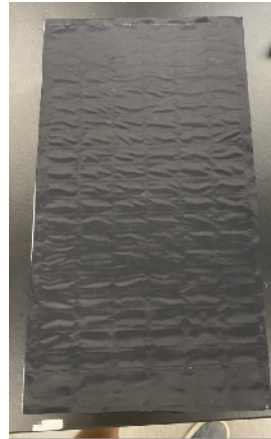


# Project Portfolio

## Spencer LeGresley

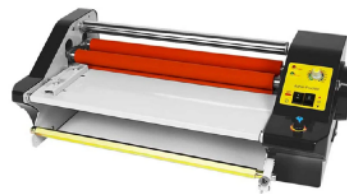
### Pouch Motor Manufacturing Machine

**Summary:** I worked under a PhD for the summer, making a machine that would produce pneumatic actuators far quicker than the previous method. I dive into challenges I faced while making the machine, and ways I sped up the process and ensured the product met our requirements. I was given total design freedom, and received only occasional input on what my PhD wanted.

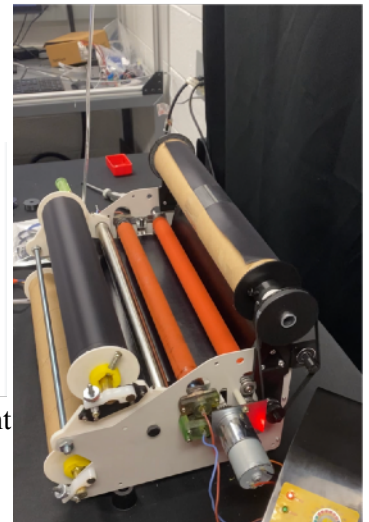


In soft robotics, pouch motors are linear actuators that are controlled with pressurized air. One type the FREE lab studies is made by pressing two plastic sheets together with a heated stamp. This manufacturing process is time consuming and low yield, and the lab needs a lot of these pouch motors for demos and experiments. I was hired to make a machine that would automate this production process. The machine was to take in two sheets of plastic, pull them through a set of turning, heated, cylindrical stamps, and spool up the finished product on a roll. I decided to start with a UV DTF laminator, which gave us a frame, heating system, and powered rollers to build on. This was faster than, and similarly priced to, developing our own setup for this generic

Uninflated and inflated pouch motors (above and above right)



Laminator before and after modification (above and right respectively)



part of the machine. I then added features that we needed that the laminator did not come with, including the custom roller, which was the most involved part of the project. The stamped roller was made with an aluminium core, and a 3D printed outer shell made of high-temperature SLA resin. The PhD student I worked for was going to make an all aluminium roller, but I suggested trying high temp resin so that I could iterate and iron out kinks in the design before spending most of the project's budget on a more expensive machined part. This let us make daily design changes, come up with simpler solutions to many of the issues we faced, and saved money by only needing to order one machined part.

Aluminium core roller with high temp resin shell.

Some of the challenges I encountered with the project were thermal expansion, rotational alignment of the top and bottom rollers, and an uneven pressure distribution between individual stamp elements.

**Thermal expansion:** the high temp resin had a lower coefficient of thermal expansion than aluminium, resulting in internal stresses that led to cracking after multiple heat cool cycles. I resolved this by slightly oversizing the resin print, so that it would fit the aluminium part properly at the operating temperature instead of room temperature. Another expansion issue was that the rods heat from an element at their centre, so there were significant temperature differences between the resin near the centre and on the outer edge, also leading to cracking. I reduced this stress by making the outer resin shell as thin as was reasonable, so that no large heat gradient would form.

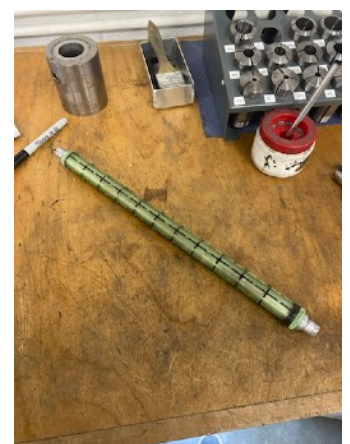
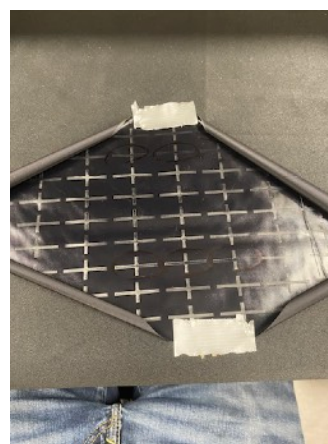


Cracking in the first iteration of the roller

**Rotational alignment:** The top roller was not driven and relied on friction to turn, and sometimes would slip relative to the bottom roller. Alignment was important to ensure consistent placement of stamps on the fabric, so I put printed gears on the outer edges of the top and bottom rollers so that they would rotate in sync. (Pictured right)

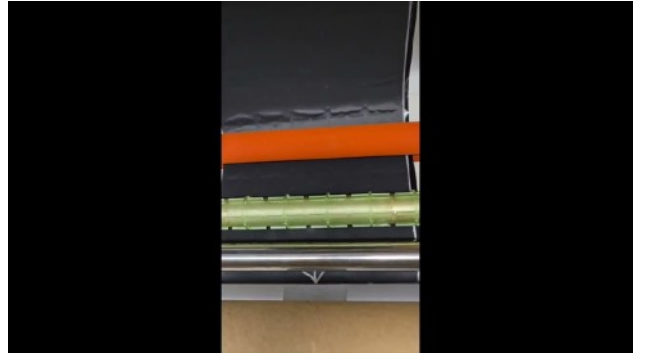


**Uneven pressure distribution:** I noticed that some of our pouches were coming unsealed at lower than expected pressures, so I ran a single sheet through the machine. I noted that some places weren't getting adequate contact/pressure judging by the appearance of the sheet (pictured right), so I tried using one rubber roller, instead of two resin ones, with the idea that the compliance of the rubber would give more flexibility on how far the profiles stuck out. I also turned down the resin roller in a lathe to ensure a consistent radius. We minimised the amount of material removed while ensuring consistent profile height by putting Sharpie on all the profiles, and turning down with very small passes until all of the black marker was removed.



The spool holders were made mostly of 3D printed parts, along with a cardboard roller and brass rods. Tensioning of the fabric was adjustable by way of a tightenable clamp on the ends of the leading spools, which would vary the torque required to make the spools rotate. The pouch motor fabric was wound up on the upper spool by

connecting the drive gears to the spool with a belt and pulley sized to match surface speeds with the stamped roller.



**Above:** Video demo of the roller machine;  
click on photo for YouTube link



## Dog-like Turkey Deterrent Robot

**Summary:** I did much of the engineering/analysis work involved in making a skid-steer that would transport a 30 lb. payload at 3 m/s, and could travel up a slope of 30°.

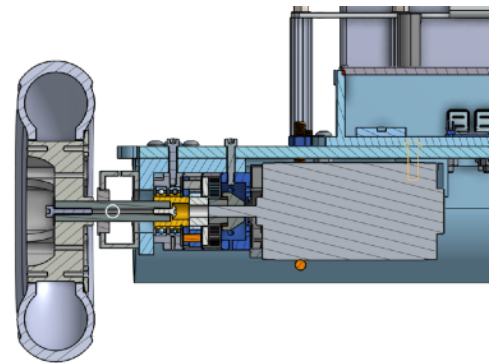
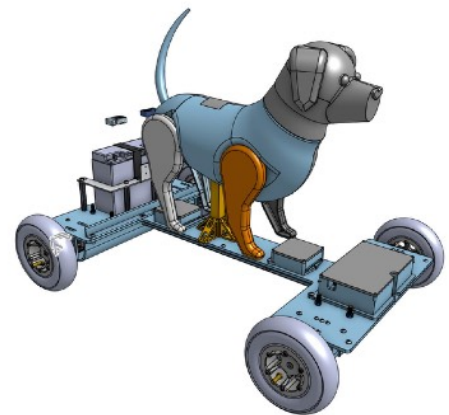
For my capstone class, my team was given the task of making a robot that would roam campus and scare off Turkeys from busy human spaces. Turkeys disrupt and scare students and damage property, particularly cars. I was on the mobility team, responsible for making the vehicle that transports the dog. The class heavily emphasised practicing a thorough design process, from problem definition all the way to verification. This involved speaking with various stakeholders, defining the problem, deciding scope, benchmarking, determining the possible set of solutions, researching solutions and selecting best ones methodically and rigorously, determining requirements and specifications, performing analysis for the design, ensuring robust integration with other subsystems as specifications changed, and ultimately verifying that our device met the requirements and specifications we originally set.

I led our group's analysis. I created a model for our operating conditions, turned those into torque-speed operating points for our motors, and ensured our motors, drivers, and battery would not exceed their current limits. I also hand-modelled stress levels in our chassis for various static loading scenarios to determine the thickness of the baseplate/chassis. Using my analysis, I selected the powertrain components, including the motor drivers and battery.

Special attention was made to ensure that load from the weight of the vehicle and its payload was not transferred to the bearings in the gearbox, as it is not made to support bending moments. This was done by supporting the axle with a bearing in a extrusion offset from the gearbox. Each powertrain assembly is mounted through a machined L extrusion that ensures all bolts have common holes locations to avoid internal stresses due to misaligned bolts. To the same end, we were generous with the size of clearance holes to avoid clashing in alignment of the outer bracket, L



**Above:** Side view of robot. Click photo for link to video of it in action



**Above:** Section view of one of the vehicle's wheels. An exploded view of the assembly is on the poster on the next page.

extrusion, gearbox, and chassis. The outer bracket was slotted, and the L extrusion was always bolted in first to allow the outer bracket to shift to align itself with the L extrusion. This was important, because misalignment between these two parts would have produced high stresses in the gearbox bearings.

I've attached our project's poster on the next page. It summarises the project at a high level and has an exploded view of the drivetrain unit.

SECTION  
INSTRUCTOR  
Dr. Shorya Awtar

ME 450 Team 18  
Spencer LeGresley, Isak Robinson, Kenneth Salinas, Gary Zheng

SPONSOR  
Dr. Jared Duquette,  
Michigan Department of  
Natural Resources

PROBLEM:

Wild turkeys are increasingly frequenting high traffic areas on the University of Michigan's North Campus.

- 56% of surveyed students reported being **intimidated** by interactions with turkeys
- 40% described interactions as **"aggressive."**



Our sponsor guided us towards an **autonomous, dog-like, turkey deterring concept** to tap into turkeys' **natural fear of dogs**. Teams were distributed as:  
Team 17 – Controls and Software  
**Team 18 – Locomotion and Power**  
Team 19 – Robotic Canine

Existing solutions

There are no existing solutions developed exclusively to deter wild turkeys



- Robotic Dogs
- High cost
  - Closed-sourced control program
  - Lacks canine appearance

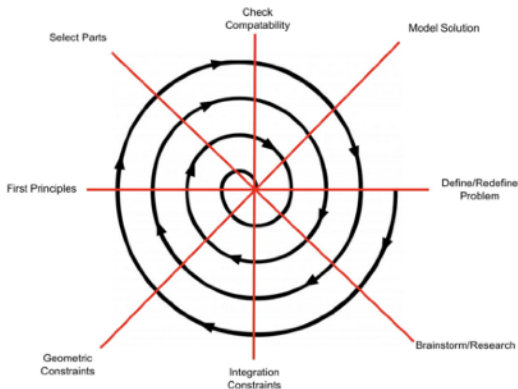


- Robot Coyote
- Requires manual control
  - Lacks canine movement

DESIGN CONTEXT

- **Dog-like movement**
- Maneuverability
- Minimally burdensome on humans
- Safety
- Autonomy
- Minimized complexity
- Potential for future application

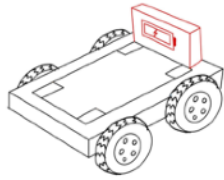
DESIGN PROCESS



REQUIREMENTS AND SPECIFICATIONS

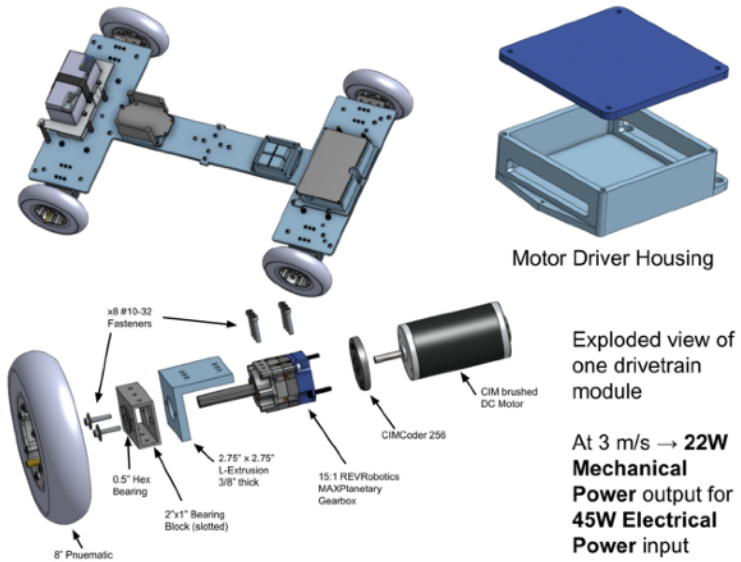
| Requirement                                | Specifications                                                | V&V methods                          | Status |
|--------------------------------------------|---------------------------------------------------------------|--------------------------------------|--------|
| Sufficient Operation Speed (High-Priority) | Instantaneous acceleration of 1.5 m/s <sup>2</sup> from 0 m/s | Acceleration trials                  | ✓      |
|                                            | Operates at 3 m/s for 5 seconds                               | Speed trials                         | ?      |
|                                            | 0 to 3 m/s and 3 to 0 m/s in ≤ 3 seconds                      | Acceleration and deceleration trials | ✓      |
| Off-Road Capability (High-Priority)        | Traverses inclination of 30° at 1 m/s                         | Steepest slope tests                 | ✓      |
|                                            | Operates on surfaces with μ = 0.15–0.35                       | Muddy surface operation              | ✓      |
| Weather Proof                              | Satisfies IEC 60529 IP44<br>Operates between -2°F to 93°F     | Test in available temperature range  | ?      |
| Sufficient Battery Life                    | 45-60 mins on 1 battery, 10% duty cycle                       | Battery Endurance Tests              | ✓      |
| Low Maintenance                            | Maintenance ≤ once per month                                  | Maintenance study                    | ?      |
| Weight                                     | Weight ≤ 60 lbs                                               | Weighing the final subsystem         | ✗      |

CONCEPT GENERATION



Tracked Design (Left) vs Wheeled Alpha Design (right)

FINAL DESIGN CONCEPT DESCRIPTION

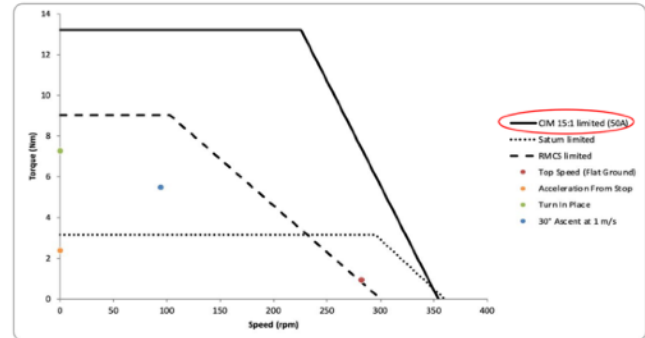


Exploded view of one drivetrain module

At 3 m/s → **22W Mechanical Power output for 45W Electrical Power input**

DESIGN ANALYSIS

Motor Selection



Torque-speed curves with torque limits for three candidate motors, and the robot's operating points.

Baseplate Material/Thickness Selection

- Aluminum with Thickness of 5/16"
- Static Chassis Analysis
  - Deflection/Stress Calculations

$$\delta = \frac{Pl^3}{48EI} \quad \sigma_{max} = \frac{Plc}{4I}$$

Battery Selection



$P_{in} = iV$   
 $P_{out} = \tau\omega$   
• 6.5 Hours of Operation at 10% Duty Cycle, 3 m/s

DESIGN CRITIQUE, DISCUSSION, AND FUTURE WORK

Strengths:

- Achieves all major objectives
- Strong maneuverability
- Strong collaboration and smooth integration

Limitations & Future Work

- Could not test temp range & snow depth
- Marginal weather resistance
- Performance optimization
- Outdoor autonomous charging
- Suspension
- Smaller footprint
- Could reevaluate a tracked vehicle

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- [4] Evans J.H., "Basic design concepts", *American Society of NavalEngineers Journal*, 71(4): 671-678, 1959
- [5] REV Robotics, "MK Battery ES17-12 - 2 Pack", *REVRobotics* <https://www.revrobotics.com/rev-19-2487>

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Dr. Shorya Awtar | Dr. Jared Duquette | ME Undergrad Machine Shop Staff | Taubman College  
FABLab Staff | Professor Art Hyde | FRC's Teams 900, 1759, & 7196 | Teams 17 & 19



## 3D printer to 3-axis CNC mill conversion

I converted a 3D printer into a CNC machine, using 3D prints made on that same printer. The conversion consisted of switching the belts for lead screws, remounting the stepper motors to accommodate the lead screws, replacing the hotend with a spindle, mounting some spoil board to put stock on, tweaking the firmware, and installing a post processor for Autodesk Fusion's gcode output. The machine worked great for wood and plastic (delrin), and after some adjustment of stepdown, engagement, feeds, and speeds, I got it working with aluminum, which is great considering how weak and low tolerance the components of this machine are. Until this point I had no experience with CAM, so making this machine was a great opportunity for me to get some experience designing with 3-axis CNC milling in mind, and for gaining an intuition for the many parameters involved when running a CNC machine.

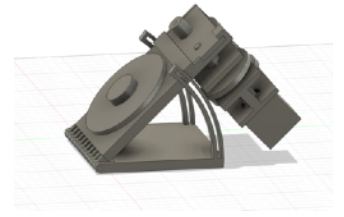
I could use this machine to make a stronger CNC machine, now that I can mill plastics and metals much stronger than 3D prints.



Video of the CNC machine in use (click for video if online)

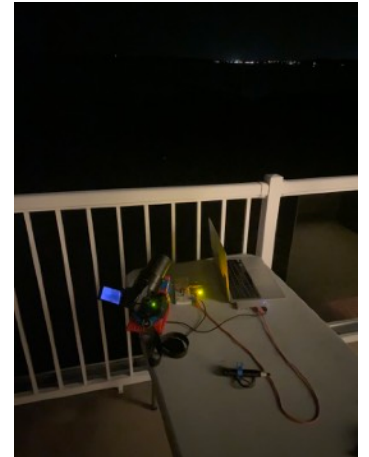
## Star Tracker

A star tracker turns a camera to negate the earth's rotation, adding clarity to photos. The device needs to rotate the camera along a parallel axis to the earth's, and at the same speed as the earth's rotation, with great precision. My star tracker achieves this at a fraction of the price of the cost of a commercial star tracker, with similar performance (70 vs 400 USD).

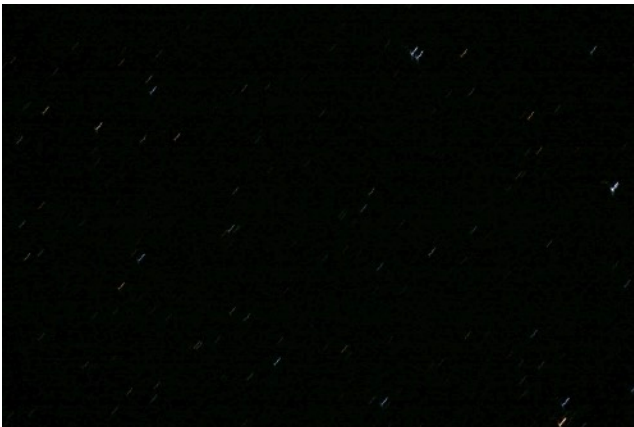


CAD design in Fusion 360

I used Fusion 360 for the 3D printed parts. They constituted the majority of the parts, such as the frame, shafts, and pulleys. Screws were fastened into the plastic parts with heat set inserts. I made an emphasis on having the assembled product have very little slop, because it needs accuracy to produce a good image. This involved incorporating adjustability into features with large tolerances, and using bearings for both radial and axial loads on the final turntable. I used a stepper motor and Arduino to drive the set of belts.



The star tracker with and without a camera mounted.



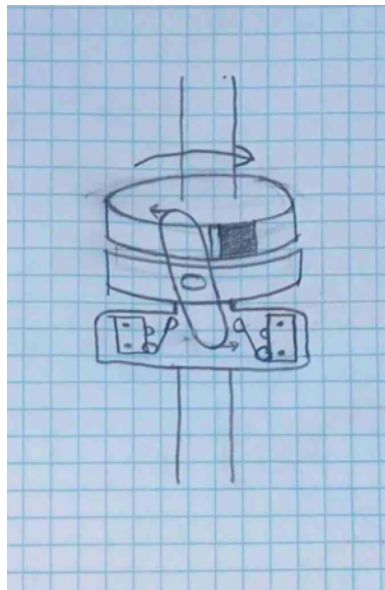
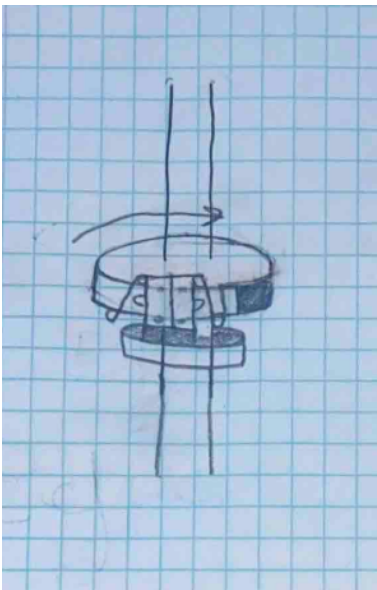
Photos before and after using the star tracker. Notice the streakiness of the stars in the left photo, and their appearance as points on the right.

Timelapse video of star tracker in action: [https://youtu.be/KiJlofM\\_mP8](https://youtu.be/KiJlofM_mP8)

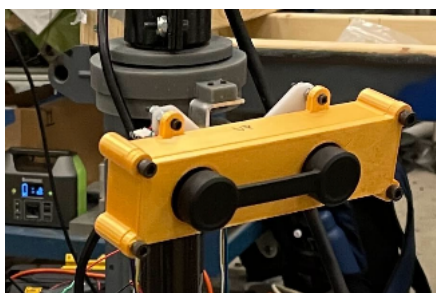
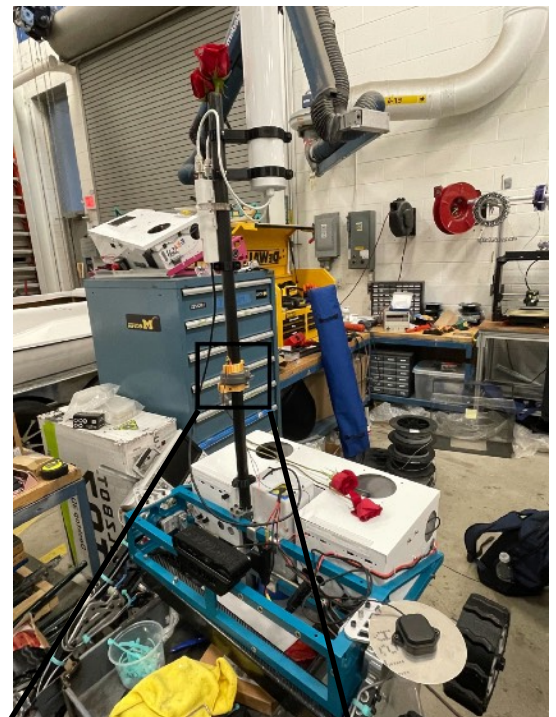


## Camera Gimbal Limit Switch Mechanism

The MROver project team Rover has a camera mounted by a two axis gimbal, and to know the exact heading of the camera, the gimbal needs a limit switch at both ends of its range of motion to define start and end angles. In the past, we have done this by having a plastic tab on the rotating part of the gimbal come into contact with the limit switch. This has limited the range of motion of the camera, because the plastic tab and the switches themselves have stopped the rotating piece from making a full 360 degree turn. I developed a mechanism to account for the thickness of the pieces, letting the gimbal rotate all the way around. Instead of the limit switches being mounted to the side of the tab, they will go below it, and be flipped by a two way switch. The two way switch lets the gimbal rotate more than 360 degrees, while still actuating the two limit switches at the end of the gimbal's range of motion. The limit switches will be mounted on the slots visible in the CAD above. The integration of the new system is a work in progress, and as such there are no photos of the final result, with limit switches installed.



Sketch of the gimbal before and after the redesign. Notice that the limit switches occupy space where the plastic tab (shaded rectangle) would otherwise go, and that the two way switch (oval) rotates out of the way of the tab to let it move further.



Camera mounted on the gimbal.



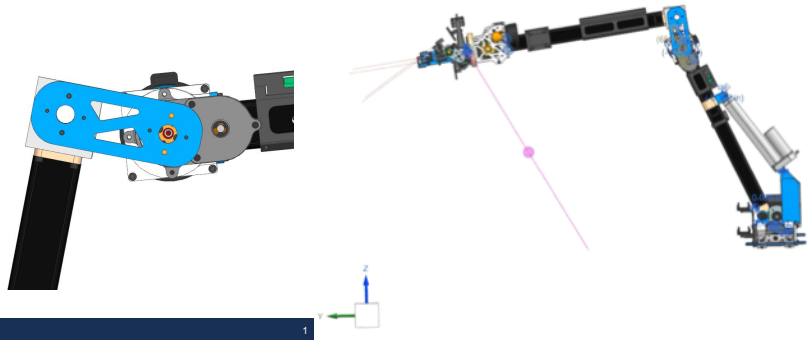
Closeup view of the gimbal relative to the entire rover.

## Robotic Arm Weight Optimization

I used Ansys to reduce the weight of a plate connecting two arm segments. The linear actuator moving the arm was underpowered, not quite moving within its range of motion within the timeframe set in the University Rover Challenge rules. Reducing this weight reduced the load on the actuator, decreasing the time for the actuator to move through its ROM by up to 5%, depending on the load on the end effector.

### Joint C

- ❖ CD Plate original mass: 1.27lbs
- ❖ CD Plate new mass: 1.03 lbs
- ❖ Mass saved 0.245 lbs
- ❖ Displacement: 0.00000103 m
- ❖ Maximum Stress: 779 kPa
- ❖ Weight savings mitigate underpowered joint B actuator

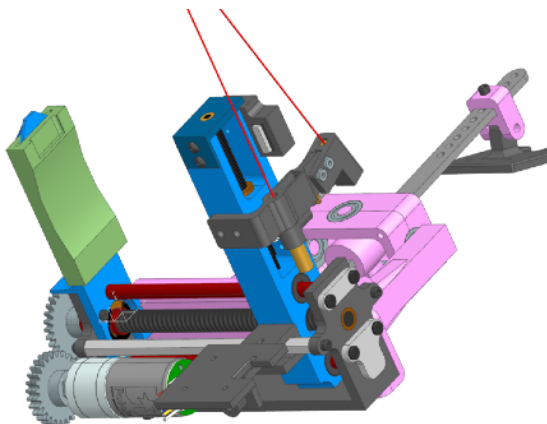


MROVER

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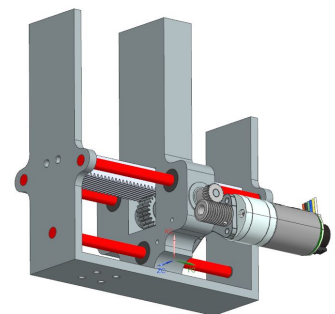
## End Effector Preliminary Redesign

I redesigned the end effector of the robotic arm in Siemens NX to prevent binding of the gripper caused by friction in the lead screw mechanism. Switching to a rack and pinion design reduces static friction in the actuator.



### EE-Alternative Design

- ❖ Rack and Pinion to mitigate lead screw binding
- ❖ Worm Gear to prevent backdrivability



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